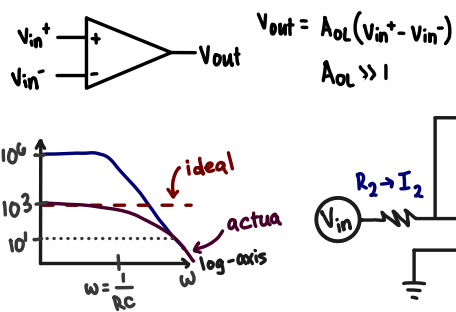


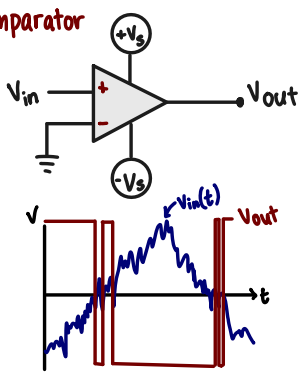
Op-amp devices and Digital Logic



Golden Rule

$V_{in}^+ = V_{in}^- = 0$ $I_1 = I_2$
 Input draws no current

Comparator



Digital Logic

- a 'bit' of information - 0 or 1 state
- $\pm 0V$ or $+5V$
 $(0-3V)$ $(3-5V)$

Logic Gate

NAND

Input	Output
A B	
0 0	1
0 1	1
1 0	1
1 1	0

NOT

Input	Output
A	
0	1
1	0

NOR

Input	Output
A B	
0 0	1
0 1	0
1 0	0
1 1	0

AND

Input	Output
A B	
0 0	0
0 1	0
1 0	0
1 1	1

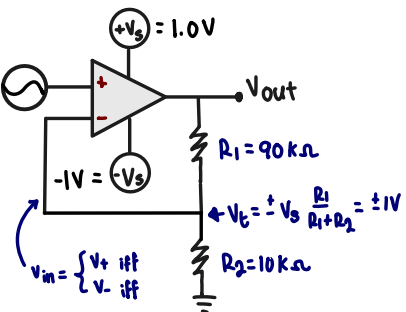
XOR

Input	Output
A B	
0 0	0
0 1	1
1 0	1
1 1	0

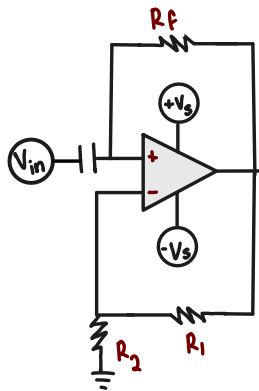
OR

Input	Output
A B	
0 0	0
0 1	1
1 0	1
1 1	1

Schmitt Trigger - Reduces Switching Behavior



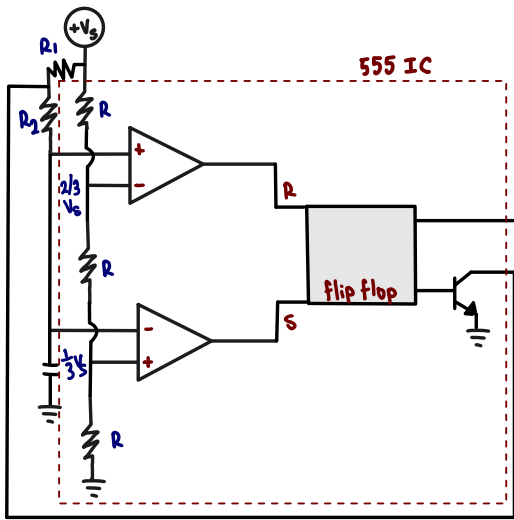
Relaxing Oscillator



Assuming Starting Point

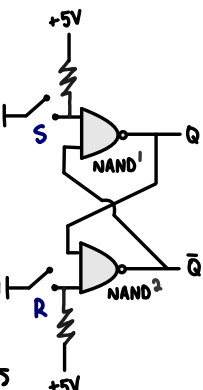
- $V_{out} = +V_s$ + C uncharged
- $V_{in}^- = 0 < V_{in}^+ = V_t = \frac{R_2}{R_1 + R_2} \cdot V_s$
- Since $V_{out} = V_s$ a current will flow through R_f + start to charge C with $\tau = R_f C$
- eventually $V_{in}^- > V_{in}^+ = V_t$
- then $V_{out} \rightarrow -V_s$
- Cap starts to discharge
- then output flips back

555 Timer Integrated Circuit



Bistable Memory

'Flip-Flop' or 'Latch'



With both switches open

- Guess: what if $Q = 0$ then $\bar{Q} = 1$
- then both inputs are high V
- Guess: what if $Q = 1$ then $\bar{Q} = 0$
- then inputs of 1 are 0 or 1 V

Input	Output
S R	Q
0 0	0
0 1	1
1 0	0
1 1	1

- now close S then $Q = 1 + \bar{Q} = 0$
- now open S nothing changes!
- now close R then $Q = 1 + \bar{Q} = 0$
- now open R nothing changes

Input	Output
S R	Q
0 0	1
0 1	1
1 0	0
1 1	0

- As long as $S + R$ are high $Q + \bar{Q}$ maintain a constant output state
- either $Q = 1, \bar{Q} = 0$ or $Q = 0, \bar{Q} = 1$

- If S goes low $\Rightarrow Q = 1, \bar{Q} = 0$
- If R goes low $\Rightarrow Q = 0, \bar{Q} = 1$

- by tethering 2 more NAND gates to input of S and R where output of NAND to S also goes to input of NAND to R, we create a D-Latch circuit

D-Latch:

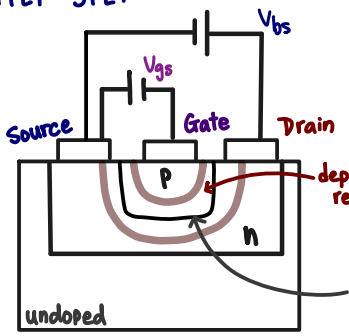
using 2 D-Latch circuits + 1 more NAND from E_1 to E_2 creates a clock

* taking this further with multiple E's wired together yields a shift register

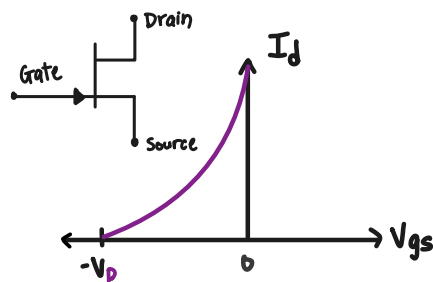
* a connected array of memory elements that can store info

Review: Field Effect Transistors

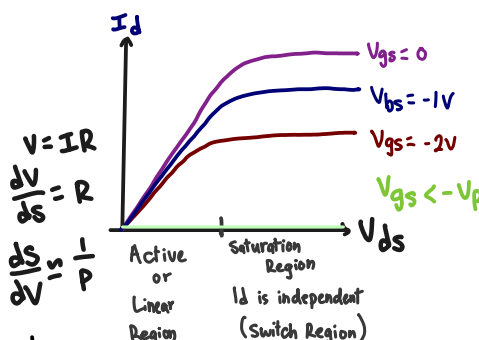
Junction FET - JFET



n-channel JFET symbol



Negative Biasing Gate makes depletion region bigger + eventually pinches off current flows from b to s



$$V = IR$$

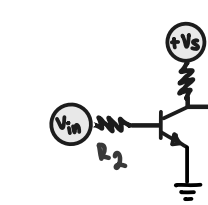
$$\frac{dV}{dS} = R$$

$$\frac{dS}{dV} = \frac{1}{R}$$

How to make Logic Gates

Data in (D) Enable (E)

Bipolar Transistor



$V_{in} = 0V$

transistor is off

$V_{out} \approx 5V$

$V_{in} = 5V$

transistor is on

choose $R_1 + R_2$

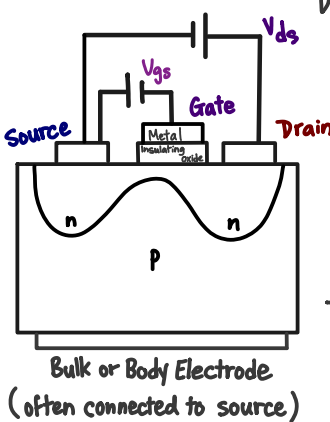
s.t. transistor is in saturation mode

$\Rightarrow$ then V_{CE} is small + $V_{out} \approx 0V$

Input	Output
D	
0	1
1	0

acts as a not gate

Metal Oxide Semiconductor FET (MOSFET)

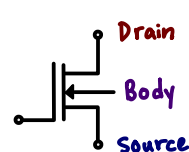


Depletion mode MOSFET

- Negative V_{gs} expands depletion region and turns off current from drain to source
- Insulating oxide layer between gate and semiconductor allows for $V_{gs} > 0$ which enhances current from Drain to Source

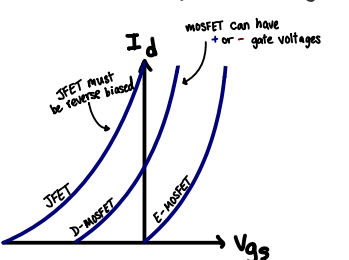
- Extra Free carriers (minority carriers in p-region) are attracted to gate + increase conductivity of the n-channel

n-channel D-MOSFET Symbol

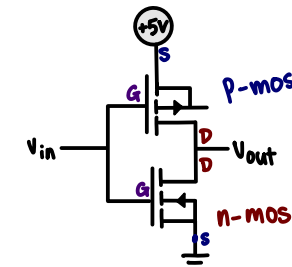


E-mos { off for $V_{gs} = 0$ on for $V_{gs} > 0$

P-mos FET { on for $V_{gs} = 0$ off for $V_{gs} > 0$



Complementary Metal Oxide Semiconductor Oxide (CMOS)



n-mos

off for $V_{gs} = 0V$

on for $V_{gs} = +5V$

for $V_{in} = +5V$

P-mos $V_{gs} = 5V$ off

n-mos $V_{gs} = 5V$ on

output is pulled to ground

P-mos

on for $V_{gs} = 0V$

off for $V_{gs} = -5V$

for $V_{in} = 0V$

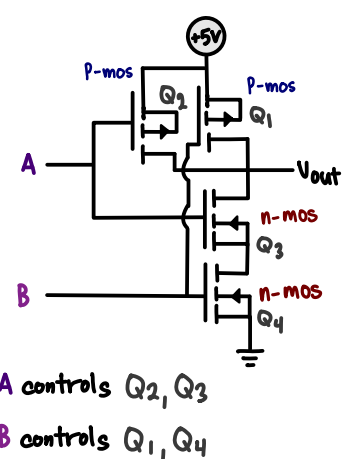
P-mos $V_{gs} = -5V$ on

n-mos $V_{gs} = 0V$ off

output is pulled to +5V

- No excess current flows from $(+5V)$ to $(\ominus)$ through the transistors
- * for either $V_{in} = 0V$ or $V_{in} = +5V$

uses much less power than NOT gate w/ 1 Transistor and 1 resistor

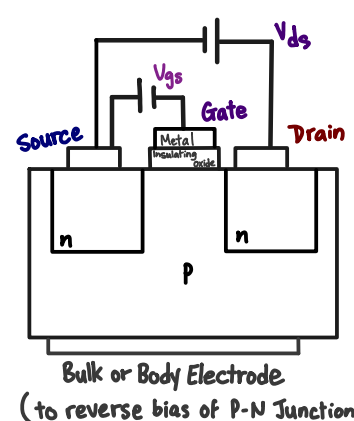


Input	Output
A B	
0 0	1
0 1	1
1 0	1
1 1	0

n-mos: $g = 1$: on $V_{gs} = V_g - V_s$ $g = 0$: off
 P-mos: $g = 0$: on $V_{gs} = V_s - V_g$ $g = 1$: off

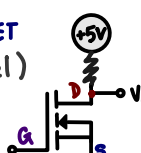
A controls Q_2, Q_3
 B controls Q_1, Q_4

Enhancement Mode MOSFET (E-MOSFET)



- If V_{gs} is positive free electrons (minority carriers in p-region) will be attracted to the gate and allow conduction from b to s

E-MOSFET (N-channel)



acts as a not gate

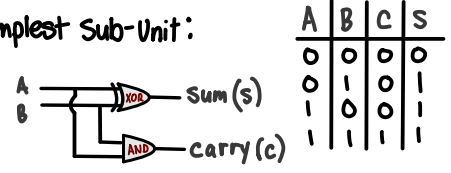
Adding Bits

Goal: Build a circuit that adds 2 numbers:

base 10	base 2	ex: 10101 = ?
0	0	
1	1	
2	10	$1 \cdot 2^4 + 0 \cdot 2^3 + 1 \cdot 2^2 + 0 \cdot 2^1 + 1 \cdot 2^0$
3	11	$= 16 + 0 + 4 + 0 + 1$
4	100	$= 21$
5	101	
6	110	
7	111	

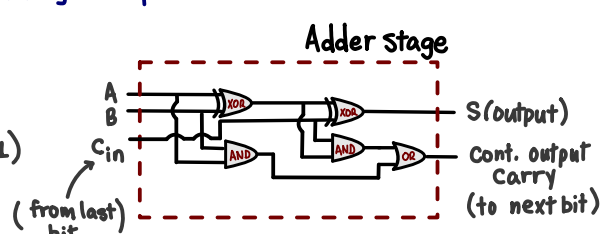
Addition: $\begin{array}{r} 3 \\ + 2 \\ \hline 5 \end{array}$

Simplest Sub-Unit:



$A + B = CS$
 least significant bit

Adding multiple Bits:



- add $A + B$ then add C_{in} at each step and check to see if we need to carry another one to the next bit!

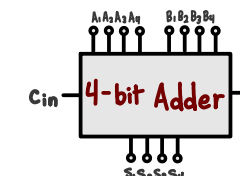
Multiple Adders

$A = A_8 A_4 A_2 A_1$ 2 Binary Numbers
 $B = B_8 B_4 B_2 B_1$ represented by 4 bits each

* $A + B = S$

$$\begin{array}{r} A_8 A_4 A_2 A_1 \\ + B_8 B_4 B_2 B_1 \\ \hline C_{8out} S_8 S_4 S_2 S_1 \end{array}$$

*



you can use two 4-bit adders to add 2 bit #'s

1 4-bit Adder = 4 1-bit Adders
 $= 4 \times 5$ logic gates
 $= 4 \times 5 \times 4$ transistors = 80 transistors

* emergent computational complexity